

ACME: Annotation-Compatible Model-Driven Exploration for Single-Source Domain Generalization in Medical Image Segmentation

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Abstract—Single-source domain generalization aims to learn medical image segmentation models from one labeled source domain that remain reliable on unseen domains. Because target-domain data are unavailable during training, existing methods commonly improve generalization by expanding source-domain variation, with recent approaches increasingly using adaptive or adversarial search to generate difficult samples. However, for dense segmentation, this strategy raises a supervision issue. A transformed image usually inherits the source annotation, even when a strong appearance change weakens the image evidence supporting that annotation. Consequently, a high-loss sample may either expose a meaningful acquisition vulnerability or provide unreliable dense supervision. We propose ACME, an Annotation-Compatible Model-Driven Exploration framework that separates these two cases. ACME learns model-sensitive directions in a bounded acquisition space and probes them at multiple strengths to identify difficult acquisition variations. The resulting candidates are then qualified using anatomical visibility, task-semantic stability, and spatial-content correspondence. Only candidates that satisfy these criteria inherit the source annotation, and the hardest admitted candidate is used for training. If no candidate is admitted, ACME reverts to the original source image. Experiments on fundus, cardiac MRI, polyp, and prostate MRI segmentation show consistent improvements over the compared domain-generalization methods across all four settings. Mechanism analysis reveals a clear trade-off between segmentation difficulty and annotation compatibility as acquisition strength increases. This trade-off motivates selecting hard samples only when the inherited supervision remains reliable.

I. INTRODUCTION

Deep neural networks have established strong baselines for medical image segmentation, from U-Net to self-configuring systems such as nnU-Net [1], [2]. Their reliability, however, often declines when deployment data differ from the training distribution, a central concern in domain generalization [3], [4]. In medical imaging, scanner hardware, imaging protocols, clinical centers, illumination, intensity response, and image quality can alter image appearance without changing the underlying segmentation target. The problem is particularly difficult in the single-source setting, where only one labeled source domain is available and target domains remain unseen during training [5], [6]. Models can therefore exploit acquisition-specific cues that are predictive on the source but unstable after deployment.

The resulting shifts are both common and heterogeneous. Cardiac bSSFP and LGE MRI can depict the same anatomy with markedly different intensity distributions, while multi-center fundus, prostate MRI, and endoscopic datasets exhibit center-, device-, and quality-dependent appearance changes.

Domain generalization addresses such shifts without target-domain training data and has been studied through invariant representation learning, meta-learning, and domain randomization [7], [8], [9]. In medical segmentation, recent single-source methods further explore gradient-guided augmentation, adversarial synthesis, and edge-guided learning [10], [11], [12].

Source-domain augmentation is therefore a natural way to create training variation. General strategies such as mixup, RandAugment, and AugMix broaden the observed data distribution [13], [14], [15]. Domain-generalization methods further perturb style, frequency content, or texture through mechanisms such as MixStyle, Fourier-based transformation, and random convolution [16], [17], [18]. Adaptive approaches go one step further by using model feedback to search for samples that expose current weaknesses [19], [20].

For dense segmentation, however, difficulty is not sufficient to determine whether an augmented sample is useful. Given a labeled source pair (x, y) , a transformed image $\tilde{x}$ usually reuses the same annotation y . This assumes that the transformed image still provides sufficient evidence for the dense label. The assumption is reasonable for mild appearance changes but becomes uncertain when the transformation is optimized to increase segmentation error. A large loss may reveal a meaningful acquisition shift, but it may also result from blurred boundaries, reduced local visibility, or altered task-relevant content. These cases can be equally difficult for the model while carrying different supervisory value. The central challenge is therefore to find difficult acquisition changes that remain compatible with the source annotation.

We address this challenge with ACME, an Annotation-Compatible Model-Driven Exploration framework for single-source domain generalization in medical image segmentation. The key idea is to separate the search for model vulnerabilities from the decision of whether a discovered failure remains suitable for supervised correction. ACME first explores a bounded acquisition space whose factors describe intensity response, contrast, channel or grayscale gain, spatial illumination, and sharpness. Exploration directions are learned from current segmentation errors and evaluated at multiple strengths, allowing the search to evolve with model capability. Candidate images are then qualified using anatomical visibility, task-semantic stability from an exponential-moving-average teacher, and spatial-content correspondence from a frozen self-supervised encoder. Candidates that fail these checks are excluded from supervision. Among those admitted, ACME uses the hardest candidate for robust training and falls back to the

source image when no candidate is suitable.

This design yields a closed Probe–Qualify–Learn process. The segmenter determines where acquisition-sensitive weaknesses remain, qualification determines which failures can inherit source supervision, and learning corrects the selected admissible failure. All auxiliary components are used only during training, so inference retains the original segmentation network without target-domain adaptation or additional forward passes. We evaluate ACME on cross-center fundus segmentation, cross-sequence cardiac MRI segmentation, multi-center prostate MRI segmentation, and cross-dataset polyp segmentation. The experiments show consistent gains across these settings and reveal a difficulty–compatibility trade-off in which stronger acquisition changes increase model difficulty while reducing the fraction of candidates suitable for source supervision.

The main contributions are summarized as follows.

- We identify annotation compatibility as an implicit condition of hard source augmentation and formulate useful hard samples as vulnerabilities that remain correctable with source supervision.
- We propose ACME, which combines model-driven acquisition exploration, candidate-level compatibility qualification, and learning from the hardest admitted candidate.
- We validate ACME across fundus, cardiac MRI, prostate MRI, and polyp segmentation, with consistent cross-domain gains and direct evidence for the trade-off between difficulty and compatibility.

II. RELATED WORK

A. Source-Domain Expansion for Medical Image Segmentation

Domain generalization seeks models that remain effective on unseen distributions without using target-domain data during training. Beyond the formulations introduced above, representative approaches use self-supervised auxiliary tasks, representation self-challenging, and style-robust features [21], [22], [23]. SWAD further improves generalization by seeking flatter solutions during training [24]. Related domain-adaptation methods instead use target-domain information, for example through adversarial alignment or correlation matching [25], [26]. Domain randomization provides another source-only route by deliberately broadening source appearance before deployment [27].

In medical segmentation, source expansion has developed from stacked image transformations toward more structured manipulation of acquisition variation. Frequency-aware randomization and dual-augmentation constraints broaden the source distribution while preserving task structure [28], [29]. StyCona decomposes style and content for augmentation, while DCCD considers both appearance and structural variation during feature disentanglement [30], [31]. These methods mainly address how to enlarge source diversity or reduce domain-sensitive representations.

A complementary line operates in frequency or feature space. Frequency-space domain randomization and federated frequency augmentation motivate structured amplitude perturbation [32], [33]. AmpMix and frequency-mixed augmentation

adapt this principle to stronger source expansion [34], [35]. Feature-statistics methods instead perturb latent styles; DSU models feature uncertainty, TriD expands medical feature distributions, and ConStyX jointly augments content and style while controlling over-augmentation [36], [37], [38]. Progressive random convolution provides another route to texture diversity [39].

ACME focuses on a different issue that becomes more important as source expansion becomes aggressive. In dense segmentation, an augmented image is normally paired with the original annotation even when the transformation weakens the image evidence supporting that label. Our question is therefore not only how much source variation to create, but which difficult variants remain suitable for supervised learning.

B. Adaptive and Adversarial Augmentation

Adaptive augmentation uses model feedback to concentrate training on informative failures rather than sampling transformations independently of the current model. In medical segmentation, AdverIN generates adversarial intensity changes through monotonic mappings and local masking [40]. More recently, TSIAA addresses over-augmentation through constrained instance-level Bézier transformations in a teacher–student adversarial framework [41]. Together with the gradient-guided and adversarial approaches introduced in the Introduction, these studies establish the value of model-aware hard augmentation.

ACME makes a further distinction between proposing a hard sample and deciding whether that sample should inherit dense supervision. Transformation constraints and consistency objectives can limit undesirable changes, but they do not explicitly make this admission decision after a candidate has been generated. ACME therefore performs candidate-level qualification before a hard sample enters supervised learning.

C. Generative and Controllable Domain Expansion

Style transfer and image translation provide broader mechanisms for changing appearance than hand-designed transforms. Neural style transfer, adaptive instance normalization, and cycle-consistent translation established widely used mechanisms for transferring appearance statistics [42], [43], [44]. Diffusion models further increased the range and controllability of image generation [45], [46]. In medical SSDG, ICMSeg combines controlled diffusion augmentation with invariant causal mechanisms for cross-modality segmentation [47]. MixStyleFlow models feature-style distributions with normalizing flows, while Latent Drifting studies controllable distribution movement for medical counterfactual generation [48], [49].

These approaches highlight a general tension between variation and preservation. Larger distributional changes can increase diversity but make task-specific constraints harder to maintain. ACME adopts progressive exploration within a restricted acquisition space rather than learning a full image distribution. More importantly, the preservation target is task specific. For dense segmentation, the relevant question is whether the transformed image still supports the annotation used to supervise it. This motivates treating model difficulty and annotation compatibility as separate properties of a candidate.

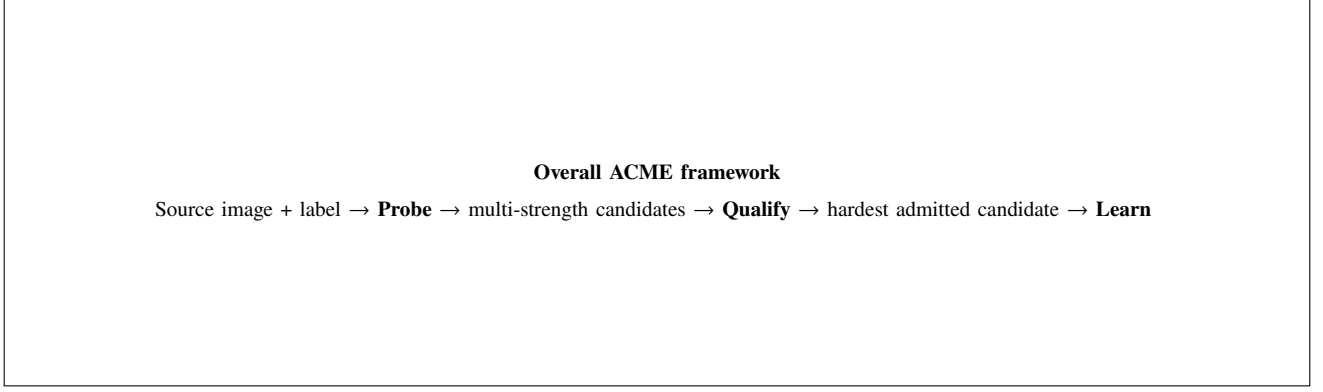


Fig. 1. Overview of ACME. Model-driven probing searches a bounded acquisition space for segmentation-sensitive variations. Annotation-compatibility qualification screens the resulting candidates using anatomical visibility, task-semantic stability, and spatial-content correspondence. The hardest admitted candidate is used for robust training together with the source image. All exploration and qualification components are removed at inference.

III. METHOD

A. Problem Formulation

Let $\mathcal{D}_s = \{(x_i, y_i)\}_{i=1}^{N_s}$ denote the labeled source domain, where $x_i \in [0, 1]^{C \times H \times W}$ is a medical image and $y_i \in \{0, \dots, C_y - 1\}^{H \times W}$ is its dense segmentation annotation. No target-domain image or annotation is available during training. The goal is to learn a segmentation model f_θ that generalizes directly to unseen acquisition domains.

Standard single-source augmentation constructs an appearance variant $\tilde{x}$ and reuses the source annotation, i.e., $(x, y) \rightarrow (\tilde{x}, y)$. This requires the transformed image to remain compatible with the dense source label. In an adversarial training setting, when augmentation is optimized to expose model failures, however, segmentation difficulty alone does not establish that this requirement still holds. We therefore separate annotation compatibility from augmentation difficulty.

B. Framework Overview

Fig. 1 summarizes the proposed Probe – Qualify – Learn framework. Given a source image x , ACME probes a bounded acquisition space using K learned directions at multiple strengths,

$$C_\theta(x) = \{G(x, \alpha \hat{d}_k) \mid k = 1, \dots, K, \alpha \in \mathcal{A}\}, \quad (1)$$

where G denotes the bounded renderer, $\hat{d}_k$ is a learned acquisition direction, and α is the exploration strength. The direction bank is optimized against the current segmenter so that probing follows its evolving vulnerabilities.

Qualification then evaluates whether the generated candidates remain suitable for inheriting the source annotation. We combine complementary evidence from anatomical visibility, task-specific prediction consistency measured by an EMA teacher, and high-level spatial-content correspondence, yielding an admissible subset $C_{\text{adm}}(x, y) \subseteq C_\theta(x)$.

Among the admitted candidates, ACME selects the hardest one,

$$\tilde{x}^* = \arg \max_{\tilde{x} \in C_{\text{adm}}(x, y)} \mathcal{H}(f_\theta(\tilde{x}), y), \quad (2)$$

where $\mathcal{H}$ measures the current segmentation difficulty. If no candidate is admitted, the source image is used instead, i.e.,

$\tilde{x}^* = x$. The segmenter is then optimized using the source image together with the selected hard candidate.

The three stages therefore serve distinct roles: *Probe* identifies model-sensitive acquisition variations, *Qualify* controls whether these variations remain eligible for dense source supervision, and *Learn* corrects the hardest admitted vulnerability.

C. Model-Driven Acquisition Vulnerability Probing

1) *Bounded Acquisition Variation*: We parameterize a compact acquisition space and let segmentation error steer exploration toward weaknesses of the current model. Specifically, the renderer takes a ten-dimensional control vector

$$z = [z_{\text{bri}}, z_{\text{con}}, z_{\text{gam}}, z_R, z_G, z_B, z_{\text{sat}}, z_{\text{hor}}, z_{\text{rad}}, z_{\text{blur}}], \quad (3)$$

covering global brightness and contrast, nonlinear intensity response, channel response, saturation, horizontal and radial illumination, and blur/sharpening. The effective parameters are bounded as

$$p(z) = s \odot \tanh(z), \quad (4)$$

where s specifies the allowable range of each factor.

The renderer applies these factors sequentially. Let $\mu(x)$ denote the global image mean and $p_{\text{rgb}} = [p_R, p_G, p_B]$. For RGB images,

$$\begin{aligned} x_1 &= (x - \mu(x))(1 + p_{\text{con}}) + \mu(x) + p_{\text{bri}} && \text{(contrast + brightness)} \\ x_2 &= \text{clip}(x_1, 10^{-4}, 1)^{\exp(p_{\text{gam}})} && \text{(gamma response)} \\ x_3 &= x_2 \odot \exp(p_{\text{rgb}}) && \text{(channel response)} \\ x_4 &= \bar{x}_3 + (x_3 - \bar{x}_3)(1 + p_{\text{sat}}) && \text{(saturation)} \end{aligned} \quad (5)$$

followed by

$$\begin{aligned} x_5 &= x_4 \odot (1 + I), && \text{(spatial illumination)} \\ x_6 &= x_5 + p_{\text{blur}} [\text{AvgPool}_{5 \times 5}(x_5) - x_5] && \text{(blur / sharpen)} \end{aligned} \quad (6)$$

Thus, positive p_{blur} produces smoothing, whereas negative values yield an unsharp-like sharpening effect. For single-channel MRI, color-specific controls are collapsed to grayscale counterparts or disabled. No geometric deformation, cropping, or spatial resampling is introduced, so all rendered variants remain pixel aligned with the source annotation. The complete renderer G denotes the composition of the above operations,

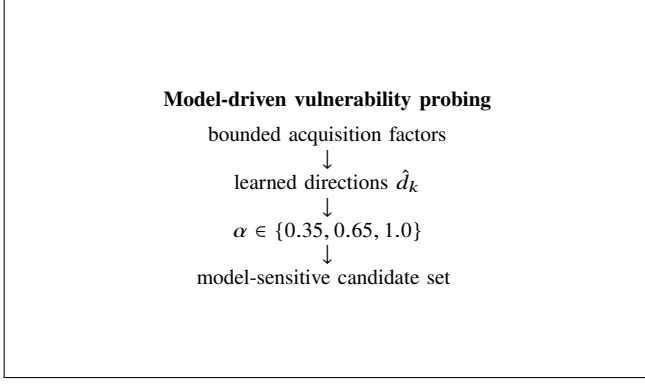


Fig. 2. Model-driven acquisition probing. Learned directions combine bounded acquisition factors according to the current segmentation error, while multiple exploration strengths examine different distances along each direction.

with $G(x, z) = x_6$; by construction, the identity condition $G(x, 0) = x$ is satisfied.

2) *Learning Model-Sensitive Directions*: ACME maintains K coupled directions in a direction bank D

$$D = [d_1, \dots, d_K]^\top, \hat{d}_k = \frac{d_k}{\|d_k\|_2}, \quad (7)$$

where each $\hat{d}_k$ specifies a coordinated change across the acquisition factors. At the exploration strength α , its segmentation difficulty is

$$\mathcal{H}_k = \mathcal{L}_{\text{Dice}}(f_\theta(G(x, \alpha \hat{d}_k)), y) + \eta \mathcal{L}_{\text{CE}}(f_\theta(G(x, \alpha \hat{d}_k)), y), \quad (8)$$

and the direction bank is optimized by

$$\max_D \frac{1}{K} \sum_{k=1}^K \mathbb{E}_{(x, y) \sim \mathcal{D}_s} [\mathcal{H}_k] - \lambda_{\text{orth}} \|\hat{D} \hat{D}^\top - I\|_F^2, \quad (9)$$

while the segmentation parameters fixed. The orthogonality penalty is designed to reduce directional collapse and maintain diversity in the direction bank. Each learned direction determines *where* to probe, while its strength α determines *how far* to move along that direction, typically using three discrete values.

D. Annotation-Compatibility Qualification

ACME evaluates annotation compatibility using three complementary criteria: anatomical visibility, task-specific prediction consistency, and spatial-content correspondence.

1) *Anatomical Visibility*: We first examine whether annotated boundaries remain supported by local image evidence. Let B_y denote foreground boundaries from y and $g(x)$ the Sobel gradient magnitude. We define

$$v(x, y) = \frac{\sum_u g(x)_u B_{y,u}}{\sum_u B_{y,u} + \epsilon}, d_{\text{anat}}(x, \tilde{x}, y) = \left| \log \frac{v(\tilde{x}, y) + \epsilon}{v(x, y) + \epsilon} \right|. \quad (10)$$

A candidate passes when $d_{\text{anat}} \leq \tau_{\text{anat}}$, thereby constraining changes to local evidence around the annotated structures.

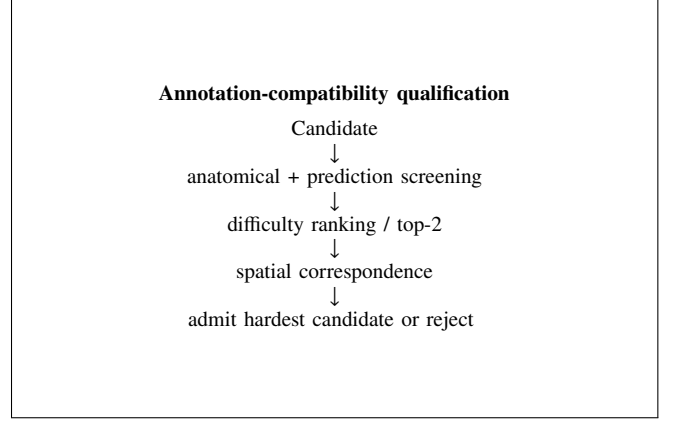


Fig. 3. Coarse-to-fine supervision admission. Anatomical visibility and task-specific prediction consistency first remove unsuitable candidates, after which spatial correspondence is verified only for the hardest survivors.

2) *Task-Specific Prediction Consistency*: To detect acquisition changes that induce excessive task-level prediction drift, ACME maintains an EMA teacher, $\bar{\theta} \leftarrow m\bar{\theta} + (1 - m)\theta$. With $p_{\bar{\theta}}(u) = \text{softmax}(f_{\bar{\theta}}(u))$, we define

$$d_{\text{pred}}(x, \tilde{x}) = \frac{1}{C_y H W} \|p_{\bar{\theta}}(x) - p_{\bar{\theta}}(\tilde{x})\|_1. \quad (11)$$

Candidates satisfying $d_{\text{pred}} \leq \tau_{\text{pred}}$ are retained. The EMA teacher provides a temporally smoothed task reference, reducing coupling between instantaneous student vulnerabilities and qualification.

3) *Spatial-Content Correspondence*: We further verify whether corresponding regions preserve similar high-level content using a frozen DINO ViT-S/8 encoder ϕ [50]. Since the renderer preserves image coordinates, corresponding patches can be directly compared:

$$d_{\text{spa}}(x, \tilde{x}) = 1 - \frac{1}{P} \sum_{p=1}^P \langle \bar{\phi}_p(x), \bar{\phi}_p(\tilde{x}) \rangle, \quad (12)$$

where $\bar{\phi}_p$ denotes the normalized feature of patch p . A candidate passes when $d_{\text{spa}} \leq \tau_{\text{spa}}$. This measure serves as a spatial-content correspondence proxy rather than a certificate of medical validity.

4) *Coarse-to-Fine Supervision Admission*: For efficiency, all candidates first undergo the anatomical and prediction-consistency checks:

$$C_{\text{cheap}} = \{\tilde{x} \in C_\theta(x) \mid d_{\text{anat}} \leq \tau_{\text{anat}}, d_{\text{pred}} \leq \tau_{\text{pred}}\}. \quad (13)$$

The survivors are ranked by segmentation difficulty, and only the two hardest are spatially verified:

$$\mathcal{T}_2 = \text{Top}_{2, \mathcal{H}}(C_{\text{cheap}}), C_{\text{adm}} = \{\tilde{x} \in \mathcal{T}_2 \mid d_{\text{spa}} \leq \tau_{\text{spa}}\}. \quad (14)$$

ACME selects the hardest member of C_{adm} according to Eq. (2); if none remains, the source image is used instead.

E. Training Objective and Optimization

Once a candidate is admitted, the student is trained jointly on the source image and the selected hard candidate. The overall objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}}(f_\theta(x), y) + \lambda_h \mathcal{L}_{\text{seg}}(f_\theta(\tilde{x}^*), y) + \lambda_c \mathcal{L}_{\text{cons}}, \quad (15)$$

where $\mathcal{L}_{\text{seg}}$ is soft Dice loss, and $\mathcal{L}_{\text{cons}}$, which predicts consistency between the source and selected hard view, is defined as

$$\mathcal{L}_{\text{cons}} = \frac{1}{HW} D_{\text{KL}}(\text{stopgrad}[p_{\theta}(x)] \| p_{\theta}(\tilde{x}^*)), \quad (16)$$

where $p_{\theta}(u) = \text{softmax}(f_{\theta}(u))$.

Training begins with a source-only warm-up to establish the segmentation model for vulnerability probing and stabilize the EMA teacher. After warm-up, optimization alternates between the direction bank and the segmenter. During direction updates, the segmentation parameters are fixed and the segmentation error is back-propagated through f_{θ} and the renderer G to update only D , while the rendering operations and their predefined ranges remain fixed. Candidate generation and qualification are then performed without gradient tracking, and the selected candidate is detached before the student update.

With D fixed, the segmenter is updated by minimizing Eq. (15), followed by the EMA teacher update. At inference, all auxiliary components are discarded and only the trained segmenter f_{θ} is retained. ACME therefore requires no target-domain data, test-time search, or additional adaptation.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

1) *Datasets and Evaluation:* We evaluate ACME on four segmentation datasets spanning different anatomies, modalities, and acquisition shifts. Labeled data from one source domain is used for training, and the unseen target sets is used for test. Table I summarizes the source and target domains and their sample sizes. Fundus-DG considers joint optic-disc and optic-cup segmentation. MSCMRSeg evaluates bidirectional generalization between C0/bSSFP and LGE while segmenting the myocardium, left ventricle, and right ventricle. The polyp task is binary lesion segmentation. The prostate task is binary whole-gland segmentation.

Dice similarity coefficient(DSC) and average surface distance(ASD) are the primary metrics, with higher DSC and lower ASD indicating better segmentation. Fundus disc and cup scores are averaged within each image and target domain, while MSCMR class scores are averaged for each reconstructed volume. Polyp ASD is computed per image in pixels, and prostate ASD is computed per reconstructed volume using the physical voxel spacing. Results are first averaged within each target domain and then macro-averaged across domains.

2) *Implementation Details:* All experiments follow the experimental protocol of StyCona for fair comparison. The segmenter is trained for 100 epochs with AdamW, a learning rate of 10^{-4} , and a batch size of 16. The random seed is fixed to 2025, and the implementation uses PyTorch 1.11.0 with CUDA 11.3. Across tasks, ACME uses the same bounded renderer and direction-bank mechanism. Spatial verification is based on a frozen DINO ViT-S/8. The complete task-level configuration is recorded in the accompanying implementation for reproducibility.

TABLE I
DATASET DOMAINS AND EVALUATION SCALES. TEST SIZES ARE TOTALS OVER THE LISTED UNSEEN TARGET DOMAINS.

Dataset	Training domain	Test domains	Training size	Test size
Fundus-DG	ORIGA	BinRushed, Drishti-GS, Magrabia, REFUGE	500 images	189 images
MSCMRSeg	C0/LGE	LGE/C0	36 subjects	45 subjects
Polyp	Kvasir	CVC-300, ColonDB, ETIS	1,450 images	636 images
Prostate	RUNMC	BIDMC, BMC, HK, I2CVB, UCL	21 cases	86 cases

B. Main Results

Tables II and III report the corresponding DSC and ASD results in the four datasets mentioned before.

On Fundus-DG, ACME obtains the highest DSC on every unseen target. Relative to StyCona, the macro DSC increases from 73.16% to 77.60%, while the macro ASD decreases from 2.63 to 2.05. The largest fundus DSC gain occurs on Magrabia at 8.92 points, while the other three centers also improve. ASD decreases on Drishti-GS, Magrabia, and REFUGE; BinRushed is the only exception, where ASD changes slightly from 3.60 to 3.70 despite the higher DSC. The gain is consistent across all four unseen centers, although the boundary metric does not improve on every individual target.

On MSCMRSeg, ACME improves both DSC and ASD in both generalization directions. For C0/bSSFP→LGE, DSC rises from 73.39% to 76.18% and ASD falls from 4.33 to 1.67. For LGE→C0/bSSFP, DSC rises from 81.59% to 83.14% and ASD falls from 2.24 to 1.64. Averaged over the two directions, ACME reaches 79.66% DSC, 2.17 points above StyCona, and reduces ASD from 3.285 to 1.659. Thus, the improvement extends from cross-center photographic variation to the substantially different intensity characteristics of cardiac MRI sequences.

For polyp segmentation, ACME achieves the highest DSC on the unseen domains, reaching a macro DSC of 68.45%. The strongest listed baseline, StyCona, obtains 60.95%, giving ACME a 7.50-point advantage. ACME also records the lowest macro ASD, 36.39, compared with 44.10 for the best competing macro result. The improvement is especially clear on CVC-300 and ColonDB, while ETIS remains the most challenging target for all methods.

For prostate MRI, ACME likewise achieves the highest DSC on all five target domains and raises the best competing macro DSC from 54.96% to 65.23%. The largest target-wise gains occur on UCL and I2CVB, where ACME reaches 71.26% and 66.36%, respectively. Its macro ASD of 6.65 is also the lowest among the compared methods, although it is not the best on every individual domain. Taken together, the results on all four settings show that ACME is not tied to a particular anatomy, image modality, or type of acquisition shift: the same training principle improves generalization in fundus photography, cardiac MRI, endoscopy, and prostate MRI.

C. Ablation Study

We conduct ablation studies on Fundus-DG to assess the contributions of the two core components in ACME: model-driven acquisition exploration and annotation-compatible supervision admission. All variants use the same evaluation protocol as the main experiment. Table IV investigates the

TABLE II
COMPARISON ON FUNDUS-DG AND MSCMRSEG. EACH CELL REPORTS DSC (%) / ASD. HIGHER DSC AND LOWER ASD ARE BETTER.

Method	Fundus-DG					MSCMRSeg	
	BinR.	Drishti	Magr.	REFUGE	Avg.	C0→LGE	LGE→C0
U-Net	54.82/7.07	77.99/1.16	61.19/8.10	79.49/1.89	68.37/4.56	65.97/6.55	76.91/3.41
AdverNoise	51.38/11.63	70.14/7.36	66.74/10.41	76.66/5.66	66.23/8.76	41.46/7.01	69.59/1.56
AdverBias	59.06/12.19	77.32/5.88	64.80/13.52	78.55/5.86	69.93/9.36	35.29/10.39	68.03/1.84
AdverHist	57.22/11.51	74.71/6.36	63.77/12.50	78.00/5.75	68.43/9.03	44.86/6.94	70.35/1.58
MixUp	63.88/10.69	75.30/6.10	67.90/10.73	79.42/5.65	71.63/8.29	41.24/7.30	69.69/1.62
RSC	53.86/12.29	69.96/7.02	67.24/8.07	75.52/5.90	66.64/8.32	39.07/8.12	72.29/1.44
AlignCE	56.05/10.95	74.08/6.48	65.16/11.27	78.14/5.84	68.36/8.63	38.29/8.32	69.93/1.56
AlignMMD	60.15/10.61	76.08/5.96	69.08/9.26	79.89/5.40	71.30/7.81	34.84/10.43	65.56/2.04
BiNorm	58.65/10.78	56.71/8.02	66.68/7.57	74.02/6.28	64.01/8.16	39.74/8.60	66.91/1.62
BNorm	39.82/16.41	44.47/8.30	49.16/17.93	72.50/7.37	51.49/12.50	39.12/7.50	69.41/1.60
CSU	62.29/10.07	65.67/7.96	67.94/7.44	77.45/5.73	68.34/7.80	46.46/5.06	69.05/1.59
DSU	59.58/11.63	66.90/7.76	69.03/8.81	77.16/5.86	68.17/8.52	40.96/7.61	69.17/1.66
INorm	37.11/4.58	12.65/8.62	45.25/10.96	22.83/4.98	29.46/7.28	58.40/6.46	67.23/1.82
pAdaIN	58.85/11.14	75.54/6.12	65.70/13.09	77.30/6.13	69.35/9.12	46.34/5.72	71.23/1.46
AmpSwap	63.56/4.36	78.16/1.13	66.13/4.45	81.84/1.45	72.41/2.85	70.93/6.48	78.74/3.26
AmpMix	65.08/4.32	77.06/1.09	68.14/4.29	79.50/1.67	72.44/2.84	63.89/6.44	80.07/2.46
FMAug	63.79/5.03	78.34/1.04	64.33/7.28	82.17/1.29	72.16/3.66	72.46/13.67	80.54/2.57
MixStyle	63.95/4.72	80.36/1.06	63.97/6.04	81.93/1.45	72.55/3.32	69.26/10.91	80.81/2.69
TriD	66.27/4.34	77.08/1.24	65.42/4.53	79.36/1.57	72.03/2.92	65.12/10.25	79.14/4.60
ConStyX	62.91/5.99	80.69/1.03	64.47/7.16	80.15/1.85	72.05/4.01	68.63/7.89	81.30/2.85
RandConv	63.75/4.70	79.35/1.13	65.61/5.07	81.52/1.46	72.56/3.09	71.90/5.65	75.17/4.27
CIDA	53.16/11.34	78.55/5.18	51.27/14.61	68.09/6.62	62.76/8.44	69.72/8.63	72.85/4.61
PRandConv	44.60/9.84	75.23/1.48	53.34/8.68	76.61/2.57	62.44/5.64	56.61/8.75	75.57/3.38
StyCona	67.80/3.60	78.43/1.08	65.71/4.37	80.68/1.49	73.16/2.63	73.39/4.33	81.59/2.24
TSIAA	58.30/9.82	73.97/7.00	70.54/9.09	78.82/5.56	70.41/7.87	64.74/1.43	83.01/1.94
ACME	69.36/3.70	82.78/0.81	74.63/2.62	83.64/1.07	77.60/2.05	76.18/1.67	83.14/1.64

TABLE III
GENERALIZATION ON UNSEEN POLYP SEGMENTATION DATASETS AND PROSTATE MRI DOMAINS. EACH ENTRY REPORTS DSC (%) / ASD.

Method	Polyp Segmentation				Prostate MRI					
	CVC-300	ColonDB	ETIS	Macro	BIDMC	BMC	HK	I2CVB	UCL	Macro
Vanilla	44.20 / 88.24	39.65 / 53.79	31.30 / 89.83	38.38 / 77.29	24.00 / 15.33	59.73 / 6.86	65.19 / 5.57	51.66 / 4.86	34.49 / 9.53	47.01 / 8.43
RandConv	30.17 / 48.31	29.25 / 47.28	23.28 / 63.76	27.57 / 53.12	35.85 / 18.60	59.73 / 8.63	59.08 / 9.76	46.52 / 11.09	45.85 / 8.15	49.41 / 11.25
MixUp	45.80 / 59.54	40.22 / 49.34	31.24 / 86.33	39.09 / 65.07	27.81 / 13.41	59.88 / 7.26	66.80 / 5.81	49.87 / 5.66	38.80 / 9.05	48.63 / 8.24
AdverNoise	39.02 / 97.31	37.80 / 59.47	31.66 / 87.28	36.16 / 81.35	24.66 / 15.88	59.48 / 7.55	66.36 / 5.67	51.72 / 4.93	35.85 / 9.39	47.61 / 8.68
AdverBias	38.12 / 100.62	34.35 / 70.22	29.08 / 108.09	33.85 / 92.98	31.44 / 9.99	63.55 / 6.28	68.64 / 5.32	49.62 / 5.28	44.36 / 8.44	51.52 / 7.06
AdverHist	37.46 / 93.97	35.02 / 57.11	28.34 / 91.04	33.61 / 80.71	23.82 / 15.05	58.53 / 8.74	66.01 / 5.57	47.80 / 5.01	42.10 / 8.64	47.65 / 8.60
RSC	48.91 / 87.58	44.29 / 71.39	27.30 / 157.42	40.17 / 105.46	19.68 / 11.94	63.46 / 5.87	61.34 / 3.98	48.25 / 5.97	35.16 / 9.66	45.58 / 7.48
MixStyle	33.11 / 149.36	38.07 / 67.57	29.30 / 76.99	33.49 / 97.97	33.54 / 16.53	61.32 / 7.67	66.22 / 6.15	54.85 / 5.91	41.61 / 8.84	51.51 / 9.02
AlignCE	37.42 / 129.83	36.38 / 78.85	26.54 / 127.89	33.45 / 112.19	25.95 / 14.04	60.87 / 6.67	67.13 / 4.65	50.43 / 4.95	35.40 / 9.57	47.96 / 7.98
AlignMMD	21.83 / 158.46	26.10 / 81.52	28.55 / 85.25	25.49 / 108.41	24.44 / 16.90	58.80 / 8.38	68.02 / 5.15	47.37 / 6.86	26.99 / 14.67	45.12 / 10.39
BiNorm	41.72 / 77.83	40.45 / 61.87	29.83 / 109.18	37.33 / 82.96	14.86 / 18.89	51.22 / 8.02	63.78 / 4.91	52.17 / 4.41	34.79 / 10.76	43.36 / 9.40
BNorm	45.73 / 73.16	40.00 / 51.66	30.72 / 97.77	38.82 / 74.20	27.66 / 15.72	60.17 / 6.63	67.64 / 5.04	52.81 / 4.78	36.37 / 9.27	48.93 / 8.29
CSU	41.68 / 68.82	39.35 / 47.76	29.38 / 62.00	36.80 / 59.53	38.74 / 16.26	62.54 / 7.24	69.78 / 5.75	52.24 / 6.29	51.49 / 7.49	54.96 / 8.61
DSU	38.18 / 105.46	36.75 / 54.97	29.32 / 56.70	34.75 / 72.38	38.65 / 14.99	61.59 / 6.91	67.84 / 6.40	52.95 / 6.69	41.12 / 8.84	52.43 / 8.77
INorm	58.20 / 24.53	43.69 / 48.12	30.69 / 59.66	44.19 / 44.10	26.17 / 18.70	54.58 / 12.50	55.71 / 9.06	48.74 / 14.02	56.02 / 9.31	48.24 / 12.72
pAdaIN	38.38 / 117.86	38.38 / 63.73	29.15 / 69.71	35.30 / 83.77	31.95 / 17.41	59.03 / 8.02	67.03 / 5.98	53.66 / 6.87	43.23 / 9.45	50.98 / 9.55
TriD	33.24 / 94.03	35.06 / 53.02	26.16 / 71.55	31.48 / 72.87	32.30 / 14.00	57.31 / 9.27	67.09 / 6.71	49.26 / 7.12	48.42 / 7.95	50.88 / 9.01
StyCona	80.13 / 29.11	60.14 / 43.78	42.57 / 89.27	60.95 / 54.05	33.54 / 16.53	61.32 / 7.67	66.22 / 6.15	54.85 / 5.91	41.61 / 8.84	51.51 / 9.02
TSIAA	60.40 / 61.12	41.63 / 121.46	25.98 / 153.43	42.67 / 112.00	30.71 / 18.20	34.62 / 22.22	28.56 / 21.68	33.12 / 21.77	41.43 / 16.77	33.69 / 20.13
ACME	85.01 / 9.34	68.55 / 32.56	51.80 / 67.28	68.45 / 36.39	43.33 / 10.81	69.70 / 5.46	75.51 / 3.63	66.36 / 5.03	71.26 / 8.32	65.23 / 6.65

effect of the learnable direction bank, while Table V evaluates the qualification mechanism and the selection of the hardest admissible candidate.

1) *Model-Driven Acquisition Exploration*: Table IV evaluates the contributions of model-driven direction learning and directional diversity to acquisition vulnerability probing. All variants use the same bounded renderer, exploration strengths, qualification procedure, and hard-sample selection rule; they differ only in how the eight probing directions are obtained and whether diversity is enforced. Exp. A freezes ACME’s initialized direction bank throughout training. Exp. B instead

resamples eight 10-dimensional random unit directions at every post-warm-up probing step, producing 24 candidates per source image with $\alpha \in \{0.35, 0.65, 1.0\}$; the directions are not cached or gradient-updated, and neither the orthogonality penalty nor QR projection is used. Exp. C enables model-driven learning after the 10-epoch warm-up, updating the directions every 8 steps from the current segmentation error with a learning rate of 0.02, but removes directional diversity by setting $\lambda_{\text{orth}} = 0$ and disabling QR projection. Standard ℓ_2 normalization is retained. Exp. D is the complete ACME exploration mechanism with

TABLE IV

ABLATION OF MODEL-DRIVEN ACQUISITION EXPLORATION ON FUNDUS-DG. “DIRECTION” SPECIFIES HOW PROBING DIRECTIONS ARE GENERATED, WHILE “DIVERSITY” INDICATES WHETHER THE DIRECTIONAL DIVERSITY CONSTRAINT IS ENABLED. EACH ENTRY REPORTS DSC (%) / ASD.

Exp.	Direction	Diversity	BinRushed	Drishti-GS	Magrabia	REFUGE	Macro
A	Fixed		64.74 / 6.43	81.11 / 0.96	71.97 / 3.32	81.82 / 1.98	74.91 / 3.17
B	Random		65.00 / 4.90	82.05 / 0.90	68.16 / 5.03	82.06 / 1.35	74.32 / 3.05
C	Model-driven		67.40 / 4.19	81.63 / 0.90	72.89 / 2.70	82.25 / 1.20	76.04 / 2.25
D	Model-driven	✓	69.36 / 3.70	82.78 / 0.81	74.63 / 2.62	83.64 / 1.07	77.60 / 2.05

TABLE V

ABLATION OF QUALIFICATION AND CANDIDATE SELECTION ON FUNDUS-DG. A CHECKMARK INDICATES ANNOTATION-COMPATIBILITY QUALIFICATION BEFORE SELECTION. EACH ENTRY REPORTS DSC (%) / ASD.

Exp.	Qualify	Selection	BinRushed	Drishti-GS	Magrabia	REFUGE	Macro
A		Random	65.59 / 10.94	82.12 / 4.66	68.76 / 10.86	80.50 / 7.95	74.24 / 8.60
B		Most Difficult	64.84 / 11.40	81.72 / 4.63	67.25 / 10.28	80.20 / 9.18	73.50 / 8.87
C	✓	Random	66.95 / 9.86	82.46 / 4.47	72.81 / 7.31	82.25 / 5.46	76.12 / 6.77
D	✓	Easiest	68.71 / 8.23	81.97 / 4.54	69.27 / 7.62	82.33 / 5.82	75.57 / 6.55
E	✓	Medium	65.88 / 10.37	82.30 / 4.44	70.58 / 9.88	80.81 / 6.30	74.89 / 7.75
F	✓	Most Difficult	69.36 / 3.70	82.78 / 0.81	74.63 / 2.62	83.64 / 1.07	77.60 / 2.05

both model-driven direction learning and directional diversity.

Without model feedback, neither fixed nor randomly resampled directions reliably identify the acquisition variations to which the evolving segmenter is vulnerable. Exp. A obtains 74.91% macro DSC and 3.17 macro ASD, while Exp. B yields a slightly lower DSC of 74.32% despite a marginal ASD reduction to 3.05. Thus, repeatedly covering the bounded acquisition space with random directions is not sufficient. When the directions are instead optimized against the current segmentation error, Exp. C improves macro DSC to 76.04% and reduces macro ASD to 2.25. Compared with Exps. A and B, this gives DSC gains of 1.13 and 1.72 points and ASD reductions of 0.92 and 0.80, respectively, demonstrating the necessity of adapting the probing directions to the model’s current vulnerabilities.

The comparison between Exps. C and D isolates the contribution of directional diversity, since both use the same model-driven update while only Exp. D enables the diversity mechanism. The complete variant further increases macro DSC from 76.04% to 77.60% and decreases macro ASD from 2.25 to 2.05, with improvements on all four target domains. This consistent gain indicates that model-driven optimization alone can produce redundant directions, whereas diversity encourages the bank to probe complementary acquisition changes. Together, these results validate both components of the proposed exploration mechanism: model feedback determines where vulnerability probing should focus, and directional diversity broadens the set of distinct vulnerabilities exposed for robust training.

2) *Qualification Before Hard-Sample Selection:* Table V evaluates the interaction between annotation-compatibility qualification and difficulty-based candidate selection. “Qualify” indicates whether candidates are first screened by the proposed annotation-compatibility criteria. When qualification is disabled, the selection rule is applied to the full candidate set; otherwise, it is applied only to the qualified subset.

Without qualification, directly favoring difficult candidates is ineffective. Exp. B selects the most difficult candidate from all

generated samples, but performs worse than random selection in Exp. A, with macro DSC decreasing from 74.24% to 73.50% and ASD increasing from 8.60 to 8.87. This indicates that difficulty alone cannot reliably identify candidates suitable for inheriting the source annotation.

After qualification, the behavior changes substantially. Exp. C already improves macro DSC to 76.12% by randomly selecting from the qualified set. Selecting the easiest or medium-difficulty candidates in Exp. D and Exp. E does not provide further gains, yielding 75.57% and 74.89% macro DSC, respectively. In contrast, Exp. F selects the most difficult qualified candidate and achieves the best result, with 77.60% macro DSC and 2.05 ASD.

The necessity of qualification is further demonstrated by directly comparing Exp. B and Exp. F, which both select the most difficult candidate but differ in whether annotation-compatibility screening is applied. Qualification improves macro DSC from 73.50% to 77.60% and reduces ASD from 8.87 to 2.05. These results show that candidate difficulty becomes beneficial only after incompatible samples are filtered out, supporting annotation-compatibility qualification as a necessary step before hard-sample selection.

D. Mechanism Analysis

Table VI and Figs. 4–5 demonstrate a single analysis of how exploration strength, segmentation difficulty, and annotation compatibility interact. We first replay the procedure of exploration and qualification with ACME’s checkpoints on Fundus-DG and MSCMRSeg’s training sets. This experiment uses the three strengths employed during training and records the actual candidate statistics and selected operating points. We further conduct a diagnostic experiment on the source validation sets with seven strengths from 0.10 to 1.00. The denser sweep tests whether the behavior observed at the three training strengths is a stable trend rather than an accidental consequence of those particular values.

The statistics in Table VI reveal a consistent strength-dependent frontier. From $\alpha = 0.35$ to 1.00, mean difficulty

TABLE VI

EXPLORATION AND QUALIFICATION STATISTICS OBTAINED BY RUNNING RETAINED CHECKPOINTS ON THE SOURCE TRAINING SETS. QUALIFIED REPORTS THE CANDIDATE-LEVEL RATE AFTER THE INEXPENSIVE CHECKS AND AFTER FINAL SPATIAL VERIFICATION. DIST. REPORTS TASK-SEMANTIC/ANATOMICAL/SPATIAL DISTANCES, AND SELECTION IS THE PROPORTION OF SELECTED CANDIDATES AT EACH STRENGTH.

Source	α	Difficulty	Qualified cheap/final	Dist. sem./anat./spa.	Selection rate
Fundus	0.35	0.1427	100.00 / 100.00	0.0040 / 0.0782 / 0.0160	3.13%
Fundus	0.65	0.1445	99.48 / 99.48	0.0065 / 0.1712 / 0.0477	21.88%
Fundus	1.00	0.1558	86.98 / 83.85	0.0098 / 0.3279 / 0.0940	75.00%
MSCMR C0	0.35	0.3098	100.00 / 100.00	0.0004 / 0.0796 / 0.0174	25.00%
MSCMR C0	0.65	0.3154	96.88 / 96.88	0.0009 / 0.1620 / 0.0470	25.00%
MSCMR C0	1.00	0.3392	80.21 / 80.21	0.0019 / 0.2803 / 0.0745	50.00%
MSCMR LGE	0.35	0.1974	100.00 / 98.96	0.0006 / 0.1177 / 0.0222	12.50%
MSCMR LGE	0.65	0.2296	91.67 / 91.67	0.0019 / 0.2843 / 0.0529	18.75%
MSCMR LGE	1.00	0.2525	73.96 / 73.96	0.0028 / 0.4834 / 0.0795	68.75%

increases from 0.1427 to 0.1558 on Fundus, from 0.3098 to 0.3392 on C0, and from 0.1974 to 0.2525 on LGE. All three compatibility distances increase at the same time. Consequently, the final qualification rate decreases from 100.00% to 83.85% on Fundus, from 100.00% to 80.21% on C0, and from 98.96% to 73.96% on LGE. Stronger exploration therefore exposes harder failures but also makes successful qualification less likely.

Importantly, qualification does not collapse the search toward easy samples. Among candidates that remain admissible, the strongest level is selected most often: 75.00% on Fundus, 50.00% on C0, and 68.75% on LGE. The mechanism thus rejects incompatible strong transformations while retaining strong transformations that still support the source annotation.

Figure 4 reproduces the same pattern across seven validation-set strengths. Difficulty and each compatibility distance rise with α , whereas the inexpensive and final qualification rates decline. This smooth behavior across the denser sweep shows that the trends observed at $\{0.35, 0.65, 1.00\}$ are not isolated measurements. The selected candidates are also weighted toward the harder end of the admissible range, particularly for Fundus and MSCMR C0. ACME therefore maintains meaningful exploration rather than satisfying compatibility simply by choosing the smallest perturbation.

However, the strength-level trends do not imply that difficulty itself can determine compatibility. Figure 5 shows substantial vertical dispersion at nearly every difficulty level: candidates with similar segmentation loss can have very different anatomical, task-semantic, or spatial distances, and both accepted and rejected candidates occur throughout the difficulty range. In other words, difficulty and compatibility distance have no strong proportional relation. A difficult sample is neither automatically incompatible nor automatically safe to supervise. This distinction explains why the explicit qualification step is necessary before selecting the most difficult candidate, and it is consistent with the ablation in Table V.

E. Comparative Analysis

Table VII compares the difficulty and annotation compatibility of the augmented samples used by each method. For adaptive methods, each sample is recorded during training, and its online

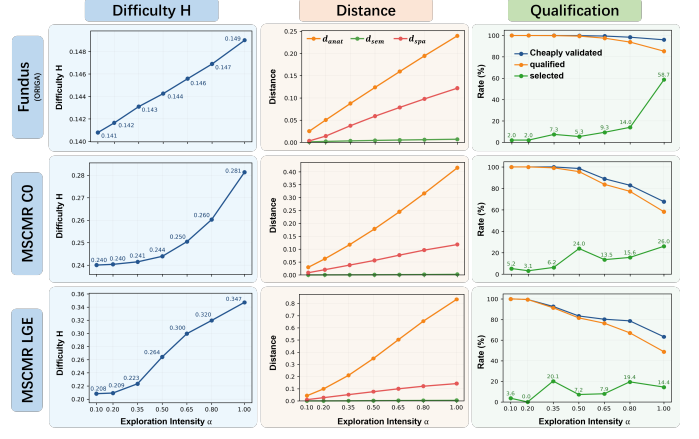


Fig. 4. Seven-level validation-set sweep of exploration strength. Rows correspond to Fundus-DG (ORIGA), MSCMR C0, and MSCMR LGE. The columns report segmentation difficulty, the three compatibility distances, and the rates passing inexpensive qualification, passing final qualification, and being selected. Increasing α consistently increases difficulty and compatibility distances while reducing qualification success; selection nevertheless favors moderate-to-strong admissible candidates.

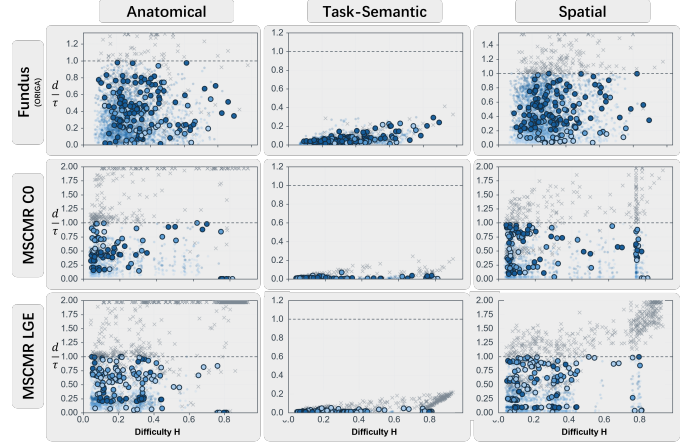


Fig. 5. Candidate-level relation between segmentation difficulty and annotation-compatibility distances on the source validation sets. Distances are normalized by their qualification thresholds; the dashed line denotes the corresponding threshold. Light points show generated candidates, crosses show candidates rejected by the full qualification procedure, and outlined circles show selected candidates. Difficulty has no strong monotonic relation with any compatibility distance, and candidates of similar difficulty can be either admissible or rejected.

TABLE VII

ONLINE HARDNESS AND COMPATIBILITY ANALYSIS ACROSS DIFFERENT TRAINING STREAMS. COMPATIBILITY IS DETERMINED BY THE PROPOSED ANATOMICAL AND SPATIAL-CONTENT CHECKS.

Method	ΔH_{online}	d_{anat}	d_{spat}	Compatible (%)	Harder $\cap$ Compat. (%)
RandConv	—	52.00	35.20	35.20	—
StyCona	—	70.00	42.40	36.00	—
TSIAA-SBF	-0.1097	41.68	61.34	29.60	4.85
TSIAA-IIA	-0.0449	35.46	27.09	11.37	2.55
AdverNoise	0.0367	91.40	68.92	68.10	63.59
AdverBias	0.0004	100.00	100.00	100.00	51.43
AdverHist	-0.0031	100.00	100.00	99.99	46.74
ACME	0.0662	100.00	100.00	99.49	97.76

difficulty change is computed immediately before optimization as $\Delta H_{\text{online},i} = H(f_{\theta_i}, x_i^{\text{aug}}, y_i) - H(f_{\theta_i}, x_i, y_i)$, where x_i and x_i^{aug} denote the input and output of the augmentation module, respectively, and H follows Eq. (8). A positive value indicates that the augmented sample is harder for the current segmenter. The SBF and IIA streams of TSIAA are reported separately. For the non-adaptive RandConv and StyCona, we apply their transformations to the ORIGA training set and report only compatibility. The proposed compatibility criteria are applied to all samples generated by competing methods, whereas the ACME statistics are computed over the samples retained by its qualification and selection procedure. Because each adaptive method is evaluated using its own evolving segmenter, ΔH_{online} should be interpreted as a within-method measure rather than an absolute comparison across methods.

The comparison reveals two common imbalances in existing augmentation methods. RandConv, StyCona, and the two TSIAA streams produce many annotation-incompatible views, with only 2.55% and 4.85% of TSIAA views being both harder and compatible. AdverNoise increases difficulty more reliably, but its jointly harder-and-compatible rate remains 63.59%. Conversely, AdverBias and AdverHist largely preserve compatibility but provide negligible mean difficulty gains. ACME achieves a better balance: its admitted views have a mean difficulty gain of 0.0662, and 97.76% are simultaneously harder and compatible. Although the methods differ in their optimization objectives, these results, together with the selection ablation, support combining model-driven vulnerability probing with explicit compatibility control.

V. DISCUSSION AND CONCLUSION

The results highlight a practical limitation of hard source-domain augmentation: samples that are difficult for the segmenter are not always suitable for supervision. As acquisition strength increases, segmentation difficulty rises together with anatomical, task-semantic, and spatial-content discrepancies, while the qualification rate decreases. This trend, together with the qualification ablations, shows that difficulty alone is insufficient for selecting training samples. ACME addresses this by separating vulnerability discovery from supervision admission. The probing stage searches for acquisition changes that expose current model weaknesses, whereas qualification determines whether the corresponding transformed image still provides sufficient evidence for the inherited dense annotation. The comparison with fixed directions further shows that adapting the search to current segmentation errors is beneficial, while directional diversity mainly helps avoid redundant exploration.

The consistent improvements on fundus photography, cardiac MRI, prostate MRI, and polyp segmentation suggest that this strategy is useful across different anatomies and acquisition shifts. Importantly, qualification does not simply favor weak perturbations. Strong candidates are frequently selected when they remain admissible, and the source image is used only when no candidate passes the criteria. The operating-point analysis therefore reflects a balance between increasing training difficulty and preserving reliable supervision. Because the renderer, direction bank, EMA teacher, candidate search, and

spatial encoder are used only during training, these gains do not introduce additional inference-time computation or require target-domain data.

Several limitations remain. The current acquisition space focuses on appearance changes, including intensity response, gain, illumination, and sharpness, and does not model geometric or other forms of domain shift. In addition, the three qualification criteria are practical indicators of annotation compatibility rather than formal guarantees, and their fixed thresholds may not transfer equally well across modalities. Candidate generation and spatial verification also increase training cost, although inference remains unchanged. Future work could extend the acquisition space and develop more adaptive admission criteria. Overall, the results suggest that hard augmentation for dense segmentation should be selected not only by how strongly it challenges the model, but also by whether the transformed image still supports the supervision used to learn from that challenge.

REFERENCES

- [1] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention*, 2015, pp. 234–241.
- [2] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, “nnu-net: a self-configuring method for deep learning-based biomedical image segmentation,” *Nature Methods*, vol. 18, pp. 203–211, 2021.
- [3] K. Zhou, Z. Liu, Y. Qiao, T. Xiang, and C. C. Loy, “Domain generalization: A survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 4, pp. 4396–4415, 2023.
- [4] I. Gulrajani and D. Lopez-Paz, “In search of lost domain generalization,” in *International Conference on Learning Representations*, 2021.
- [5] L. Zhang, X. Wang, D. Yang, T. Sanford, S. Harmon, B. Turkbey, B. J. Wood, H. Roth, A. Myronenko, D. Xu, and Z. Xu, “Generalizing deep learning for medical image segmentation to unseen domains via deep stacked transformation,” *IEEE Transactions on Medical Imaging*, vol. 39, no. 7, pp. 2531–2540, 2020.
- [6] C. Ouyang, C. Chen, S. Li, Z. Li, C. Qin, W. Bai, and D. Rueckert, “Causality-inspired single-source domain generalization for medical image segmentation,” *IEEE Transactions on Medical Imaging*, vol. 42, no. 4, pp. 1095–1106, 2023.
- [7] K. Muandet, D. Balduzzi, and B. Schölkopf, “Domain generalization via invariant feature representation,” in *International Conference on Machine Learning*, 2013, pp. 10–18.
- [8] D. Li, Y. Yang, Y.-Z. Song, and T. M. Hospedales, “Learning to generalize: Meta-learning for domain generalization,” in *AAAI Conference on Artificial Intelligence*, 2018.
- [9] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, “Domain randomization for transferring deep neural networks from simulation to the real world,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2017, pp. 23–30.
- [10] Z. Su, K. Yao, X. Yang, K. Huang, Q. Wang, and J. Sun, “Rethinking data augmentation for single-source domain generalization in medical image segmentation,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 2, 2023.
- [11] Y. Xu, S. Xie, M. Reynolds, M. Ragoza, M. Gong, and K. Batmanghelich, “Adversarial consistency for single domain generalization in medical image segmentation,” in *Medical Image Computing and Computer Assisted Intervention*, vol. 13437, 2022, pp. 671–681.
- [12] J. Jiang and S. Gu, “Train once, deploy anywhere: Edge-guided single-source domain generalization for medical image segmentation,” in *Medical Imaging with Deep Learning*, vol. 250, 2024, pp. 722–741.
- [13] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, “mixup: Beyond empirical risk minimization,” in *International Conference on Learning Representations*, 2018.
- [14] E. D. Cubuk, B. Zoph, J. Shlens, and Q. V. Le, “Randaugment: Practical automated data augmentation with a reduced search space,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2020.

- [15] D. Hendrycks, N. Mu, E. D. Cubuk, B. Zoph, J. Gilmer, and B. Lakshminarayanan, "Augmix: A simple data processing method to improve robustness and uncertainty," in *International Conference on Learning Representations*, 2020.
- [16] K. Zhou, Y. Yang, Y. Qiao, and T. Xiang, "Domain generalization with mixstyle," in *International Conference on Learning Representations*, 2021.
- [17] Y. Yang and S. Soatto, "Fda: Fourier domain adaptation for semantic segmentation," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020.
- [18] Z. Xu, D. Liu, J. Yang, C. Raffel, and M. Niethammer, "Robust and generalizable visual representation learning via random convolutions," in *International Conference on Learning Representations*, 2021.
- [19] S. Shankar, V. Piratla, S. Chakrabarti, S. Chaudhuri, P. Jyothis, and S. Sarawagi, "Generalizing across domains via cross-gradient training," in *International Conference on Learning Representations*, 2018.
- [20] R. Volpi, H. Namkoong, O. Sener, J. Duchi, V. Murino, and S. Savarese, "Generalizing to unseen domains via adversarial data augmentation," in *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [21] F. M. Carlucci, A. D'Innocente, S. Bucci, B. Caputo, and T. Tommasi, "Domain generalization by solving jigsaw puzzles," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.
- [22] Z. Huang, H. Wang, E. P. Xing, and D. Huang, "Self-challenging improves cross-domain generalization," in *European Conference on Computer Vision*, 2020.
- [23] H. Nam, H. Lee, J. Park, W. Yoon, and D. Yoo, "Reducing domain gap by reducing style bias," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021.
- [24] J. Cha, S. Chun, K. Lee, H.-C. Cho, S. Park, Y. Lee, and S. Park, "Swad: Domain generalization by seeking flat minima," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 22 405–22 418.
- [25] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, "Domain-adversarial training of neural networks," *Journal of Machine Learning Research*, vol. 17, no. 59, pp. 1–35, 2016.
- [26] B. Sun and K. Saenko, "Deep coral: Correlation alignment for deep domain adaptation," in *ECCV Workshops*, 2016.
- [27] X. Yue, Y. Zhang, S. Zhao, A. Sangiovanni-Vincentelli, K. Keutzer, and B. Gong, "Domain randomization and pyramid consistency: Simulation-to-real generalization without accessing target domain data," in *IEEE/CVF International Conference on Computer Vision*, 2019, pp. 2100–2110.
- [28] J. Wu, Z. Chen, Y. Chen, Z. Cui, S. Du, X. Yang, and J. Xin, "Frequency-aware domain randomization for single-source domain generalization in medical image segmentation," *Medical Physics*, vol. 52, no. 12, p. e70118, 2025.
- [29] R. Wang, J. Guo, J. Zhang, L. Qi, Q. Yu, and Y. Shi, "A hybrid dual-augmentation constraint framework for single-source domain generalization in medical image segmentation," *Pattern Recognition*, vol. 170, p. 112082, 2026.
- [30] Z. Shen, P. Cao, J. Yang, O. R. Zaiane, and Z. Chen, "Style content decomposition-based data augmentation for domain generalizable medical image segmentation," in *Medical Imaging with Deep Learning*, vol. 315, 2026, pp. 21–35.
- [31] Y. Li, B. Ren, Y. Ren, H. Chen, M. Wang, and Z. Liu, "Dccd: Dual-contrastive channel disentanglement for single-source domain generalization in medical image segmentation," *Neurocomputing*, vol. 700, p. 134458, 2026.
- [32] J. Huang, D. Guan, A. Xiao, and S. Lu, "Fsdr: Frequency space domain randomization for domain generalization," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 6891–6902.
- [33] Q. Liu, C. Chen, J. Qin, Q. Dou, and P.-A. Heng, "Feddg: Federated domain generalization on medical image segmentation via episodic learning in continuous frequency space," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 1013–1023.
- [34] Q. Xu, R. Zhang, Z. Fan, Y. Wang, Y.-Y. Wu, and Y. Zhang, "Fourier-based augmentation with applications to domain generalization," *Pattern Recognition*, vol. 139, p. 109474, 2023.
- [35] H. Li, H. Li, W. Zhao, H. Fu, X. Su, Y. Hu, and J. Liu, "Frequency-mixed single-source domain generalization for medical image segmentation," in *Medical Image Computing and Computer Assisted Intervention*, 2023, pp. 127–136.
- [36] X. Li, Y. Dai, Y. Ge, J. Liu, Y. Shan, and L.-Y. Duan, "Uncertainty modeling for out-of-distribution generalization," in *International Conference on Learning Representations*, 2022.
- [37] Z. Chen, Y. Pan, Y. Ye, H. Cui, and Y. Xia, "Treasure in distribution: A domain randomization based multi-source domain generalization for 2d medical image segmentation," in *Medical Image Computing and Computer Assisted Intervention*, 2023.
- [38] X. Chen, Z. Shen, P. Cao, J. Yang, and O. R. Zaiane, "Constyle: Content style augmentation for generalizable medical image segmentation," *arXiv preprint arXiv:2506.10675*, 2025.
- [39] S. Choi, D. Das, S. Choi, S. Yang, H. Park, and S. Yun, "Progressive random convolutions for single domain generalization," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- [40] Z. Zhang, B. Wang, L. Yao, U. Demir, D. Jha, I. B. Turkbey, B. Gong, and U. Bagci, "Domain generalization with adversarial intensity attack for medical image segmentation," *arXiv preprint arXiv:2304.02720*, 2023.
- [41] Z. Wang, L. Chen, X. Xie, Y. Zhang, Y. Cai, and W. Ding, "Teacher-student instance-level adversarial augmentation for single domain generalized medical image segmentation," *IEEE Transactions on Medical Imaging*, 2026.
- [42] L. A. Gatys, A. S. Ecker, and M. Bethge, "Image style transfer using convolutional neural networks," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 2414–2423.
- [43] X. Huang and S. Belongie, "Arbitrary style transfer in real-time with adaptive instance normalization," in *IEEE International Conference on Computer Vision*, 2017, pp. 1501–1510.
- [44] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *IEEE International Conference on Computer Vision*, 2017, pp. 2223–2232.
- [45] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851.
- [46] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 10 684–10 695.
- [47] B. Chen, Y. Zhu, Y. Ao, S. Caprara, R. Sutter, G. R  tsch, E. Konukoglu, and A. Susmelj, "Generalizable single-source cross-modality medical image segmentation via invariant causal mechanisms," in *IEEE/CVF Winter Conference on Applications of Computer Vision*, 2025, pp. 3592–3602.
- [48] R. Safdari, M.-A. N. Mahani, M. Koohi-Moghadam, and K. T. Bae, "Mixstyleflow: Domain generalization in medical image segmentation using normalizing flows," in *Medical Image Computing and Computer Assisted Intervention*, 2025.
- [49] Y. Yeganeh, I. Charisiadis, M. Hasny, M. Hartenberger, B. Ommer, N. Navab, A. Farshad, and E. Adeli, "Latent drifting in diffusion models for counterfactual medical image synthesis," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [50] M. Caron, H. Touvron, I. Misra, H. J  gou, J. Mairal, P. Bojanowski, and A. Joulin, "Emerging properties in self-supervised vision transformers," in *IEEE/CVF International Conference on Computer Vision*, 2021, pp. 9650–9660.